Laboratories

1. **Mechanical & Civil Engineering Workshop** (Faculty in Charge: Dr. Binulal B R, Associate Professor)

The Workshops include elementary and basic concepts of the practical side of Mechanical Engineering. Different sections of mechanical workshop are:

- Fitting section
- Sheet metal section
- Carpentry Section
- Welding Section

In this basic engineering workshop, the students acquire the knowledge of various trades and the usage of different types of tools like files, hammers, chisels, snipers, anvils, cutting pliers, welding rods and different kinds of materials.

The Civil Engineering workshop provides basic ideas about surveying and leveling, setting out buildings and different types of brick bonds.

2. **Material Testing Lab** (Faculty in Charge: Mr. Santhosh T, Assistant Professor)

This Laboratory offers facilities for testing building materials for their strength, behavior and suitability for various applications. Experiments include tension tests, compression tests, torsion tests, shear tests, beam bending, and column buckling. Following are the main machines/equipment in this lab:

- Universal Testing Machine
- Torsion Testing Machine
- Brinell & Rockwell Hardness Tester
- Impact Testing Machine
- Spring Testing Machine

Hardness Tester

Spring Testing Machine

Impact Testing Machine

Torsion Testing Machine

Universal Testing Machine

Understanding of various materials and its Mechanical Properties can be tested by the students and with the equipment placed in pleasant atmosphere of the laboratory. The strength of a given specimen is analyzed and its relative stress and strain diagram is studied.

3. **Fluid Mechanics & Machines Lab** (Faculty in Charge: Mr. Madhu A K, Assistant Professor)

Fluid Mechanics lab has ample facilities for conducting fluid flow experiments. The lab is equipped with different flow measuring set ups such as venturimeter, orifice-plate, Pitot tube etc., where students can visualize the basic theory of working of the flow meter. The lab also has Reynolds's apparatus, flow in open channels and set up to measure the impact of jet on stationary vanes. The lab helps the students to consolidate the concepts learned in the theory course in Fluid Mechanics.

The lab is equipped with the following experiments:

- Impact of Jet
- Bernoulli's Theorem Verification
- Reynolds Experiment of Flow Determination
- Major (Friction) Losses in Pipe Flow
- Minor Losses in Pipe Flow
- Velocity Measurement Using Pitot - static tube
- Measurements of Flow through Pipe Using Venturimeter
- Measurements of Flow through Pipe Using Nozzle meter
- Measurements of Flow through Pipe Using Orifice meter
- Open Channel Flow Measurements Using Notches
- Hydraulic Coefficients of Orifice
- Flow through Mouthpiece
- Metacentric Height of Floating Bodies

Flow measurement experimental setups

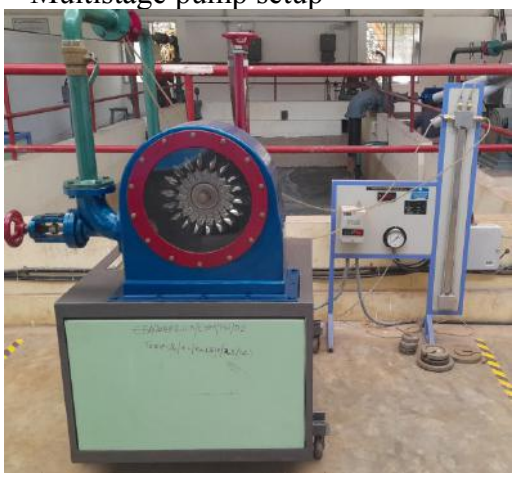
Fluid Machines lab is particularly established to study the performance of hydraulic machines like pumps and turbines. The students can observe the change of hydraulic power into mechanical or electrical energy in case of turbines and the conversion of electrical energy into hydraulic power in centrifugal pumps. The main experimental setups in the fluid machines lab are:

- Pelton Wheel Turbine
- Francis Turbine Test Rig
- Kaplan Turbine Test Rig
- Variable Speed Centrifugal Pump Test Rig
- Rated Speed Centrifugal Pump Test Rig
- Multi Stage Centrifugal Pump Test Rig
- Reciprocating Pump Test Rig
- Gear Pump Test Rig
- Jet Pump Test Rig
- Submersible Pump Test Rig
- Hydraulic Ram Test Rig

Hydraulic pump experimental setups

Multistage pump setup

Hydraulic ram



Hydraulic turbine test rigs

4. **Thermal Engineering Lab** (Faculty in Charge : Mr. Revikumar Thampi V R, Assistant Professor)

The objective of the thermal engineering laboratory is to introduce the student to the fundamental theories and the industrial applications of thermodynamics, heat transfer, and fluid mechanics. The Thermal Engineering Laboratory is equipped with several rigs and test facilities for doing research within thermal processes for energy conversion. Equipments available in this laboratory are:

- Rotary Air Compressor
- Two Stage 3 HP Reciprocating Air Compressor
- Four Stroke Diesel Engine with Electrical resistance Loading system
- Four Stroke Diesel Engine with Brake Drum Loading system
- 4 Stroke 4 Cylinder Petrol Engine with Brake Drum Loading system
- Air-conditioning and Refrigeration Tutor
- Valve Timing Model of Four Stroke Diesel Engine
- Valve Timing Model of Two Stroke Petrol Engine
- Flash and Fire point Apparatus
- Redwood Viscometer
- Junkers Gas Calorimeter

Vapor compression refrigeration setup

Vapor absorption refrigeration setup

Centrifugal compressor setup

Reciprocating compressor setup

Rotary compressor setup

Air-conditioning setup

Air conditioning setup

IC Engine test setup

IC engine setup

5. **Heat and Mass Transfer Lab** (Faculty in Charge: Mr. Arun Kumar C, Assistant Professor)

Heat and mass transfer are the areas of engineering that address how quickly energy or material substances move from one place to another. Understanding these phenomena is essential to designing and analyzing heating and cooling operations, and most separation processes. The equipments in this laboratory are

- Shell and Tube Heat Exchanger
- Parallel flow/Counter flow Heat Exchanger
- Emissivity Measurement Apparatus
- Thermal Conductivity of a Metal Rod Apparatus
- Unsteady State Heat Transfer Apparatus
- Drop wise and Film wise Condensation Apparatus
- Boiling Heat Transfer Apparatus
- Critical Heat Flux Apparatus
- Forced Convection Heat Transfer Apparatus

Heat transfer experimental setups

Heat transfer experimental setups

6. **Mechanical Engineering Lab** (Faculty in Charge: Dr. Jose Mathew, Associate Professor)

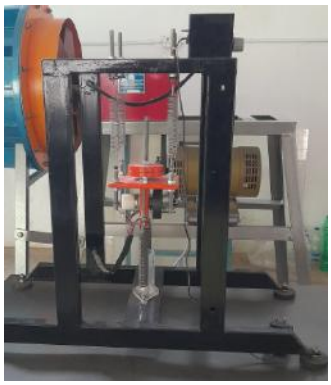
The course is intended to enable the students to get an exposure to equipment related to machine dynamics, cutting forces in milling machine, basics of pneumatic and hydraulic devices, basic concepts of stepper motors, basic ideas of data acquisition systems and automation. The equipment in the Mechanical Engineering lab are:

- Gyroscope apparatus
- Universal governor apparatus
- Whirling of shaft apparatus
- Free and forced vibration apparatus
- Stepper motor trainer
- Servo motor trainer
- Pneumatic trainer with PLC controller
- NDT setup

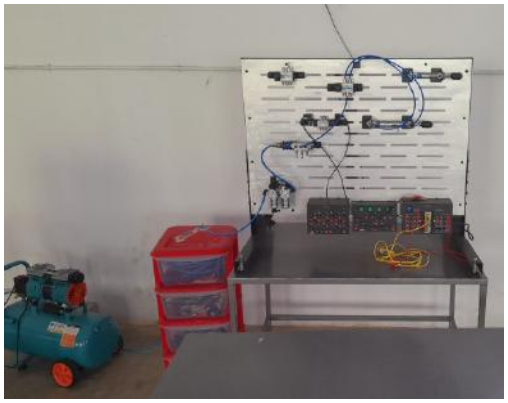
Whirling of shaft apparatus

Universal governor apparatus

Gyroscope apparatus



Free & forced vibration apparatus



Pneumatic trainer with PLC controller

7. **Metrology & Measurements Lab** (Faculty in Charge: Faculty in Charge: Mr. Baiju V, Assistant Professor)

The accuracy and correct measurement improves the quality of the product. To maintain the deserved quality the engineers must have the skills and technology regarding measurement, measuring tools and measuring procedures. This lab provides a large number of measuring tools of great accuracy. This lab provides the ideas and techniques about the correct measurement of linear, angular measurement. The student can learn the suitable techniques to find out the errors in measured quantities by using the modern measuring instruments.

The instruments and equipments in this lab are

- Tool Maker's Microscope
- Ultrasonic Flaw Detector
- Thermocouple Calibration Test Rig
- Profile Projector
- Milling Tool Dynamometer
- Lathe Tool Dynamometer
- Instruments or linear and angular measurements

Profile projector

Surface roughness tester

Surface plate

8. **Machine shop** (Faculty in Charge: Mr. Baiju V, Assistant Professor)

In Machine shop students modify their working skills and study metal cutting principles in machine tools like Lathe, Drilling machine etc. and develop their skills on work models to learn operations like milling, slotting, shaping grinding and planning in machine tools like, Milling machines, Slotting machines, Shaping machines, Surface grinding machines and Planning machines. The list of machines in the machine shop is as given below.

- Lathe
- Slotting Machine
- Planing Machine
- Surface Grinding Machine
- Shaper
- Universal Milling Machine
- Vertical Turret Milling Machine
- Horizontal Milling Machine
- Cylindrical Grinding Machine
- Radial Drilling Machine
- Pedestal Drilling Machine
- Hydraulic Hacksaw
- Power Hacksaw

Lathe

Lathe

Cylindrical grinder

Surface grinder

Universal milling machine

Tool & cutter grinder

Vertical milling machine

Horizontal milling machine

Drilling machines

Slotter machine

Shaper machine

Planer machine

9. CAD/CAM Lab (Faculty in Charge: Dr. Venkitaraj K P, Assistant Professor)

The CAD/CAM lab aims to support teaching and research in the areas of innovative design and intelligent manufacturing. The lab provides advanced design and manufacturing theory and practice environments which fit modern industry requirements. The department has a well-equipped computer laboratory with advanced softwares such as AUTODESK INVENTOR, SOLID EDGE, ANSYS, HYPER WORKS, PRIMAVERA etc. Students are also made familiar with modern equipment like CNC trainer lathe, Robotic arm trainer etc.

CAD Lab

HMT CNC Trainer

10. CFD Lab (Faculty in Charge: Dr. Jose Mathew, Associate Professor)
This lab provides the computational facility for doing the project work.

CFD lab

High performance server

11. Advanced Thermal & Fluid Research Lab (Faculty in Charge: Dr. Venkitaraj K P, Assistant Professor)

This lab enables the faculty & students to undertake various research project works in the area of thermal and fluid flow. The facilities in the lab includes a low speed wind tunnel test rig, multi-fuel variable compression ratio engine test rig, rotational viscometer, temperature bath, ultrasonic shaker, data acquisition system, digital storage oscilloscope electronic weighing balance, temperature calibrator, muffle furnace, hot air oven, laboratory ball mill, magnetic stirrer, pH meter, hot plate etc.

Low Speed Wind Tunnel Test Rig

Multi-fuel VCR Engine Test Rig

Brookfield Rotational Viscometer

Brookfield Temperature Bath

Ultrasonic Shaker

Data Acquisition System (Keysight)

Infrared Thermal Imager (Fluke/Flir)

Fluke Drywell Calibrator

Muffle Furnace

12. Department Library (Faculty in Charge: Mr. Thoufeeq R, Assistant Professor)

In addition to the Central Library, a well-equipped department library is functioning in the department catering to the needs of the faculty and students. The department library contains texts & reference books, seminar reports, project reports, previous year question papers etc.

Department library

13. Seminar Hall (Faculty in Charge: Dr. Venkitaraj K P, Assistant Professor

The department has a seminar hall equipped with a video conferencing facility and laptops.

Seminar hall